ASSIGNMENT 10.2

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BATCH-30

Task:
Use AI to analyze a Python script and correct all syntax and logical errors.
Sample Input Code:
def calculate_total(nums)
sum = 0
for n in nums
sum += n
return total
Expected Output-1:
Corrected and executable Python code with brief explanations of the identified syntax and logic errors

CODE:

```
def calculate_total(nums):
    total = 0
    for n in nums:
        total += n
    return total

# Example usage
numbers = [1, 2, 3, 4, 5]
print(calculate_total(numbers))]
```

OUTPUT:

```
PS D:\AI Assisted coding>  & 'C:\Users\vinee\AppData\Local\Microsoft\WindowsApps\python3.12.exe' 'c:\Users\vinee\.vscode\extensions\ms-python.debugpy-2025.19.202602
2301-win32-x64\bundled\libs\debugpy\launcher' '64969' '--' 'D:\AI Assisted coding\10.2-1.py'
15
PS D:\AI Assisted coding>
```

Task Description -2(Code Style Standardization)
Task:
Use AI to refactor Python code to comply with standard coding style guidelines.
Sample Input Code:
def findSum(a,b):return a+b
print(findSum(5,10))
Expected Output-2:
Well-structured, consistently formatted Python code following standard style conventions.

CODE:

```
def findSum(a,b):
    return a+b

print(findSum(5,10))
```

#give what was the error before in the above code and how it was fixedThe error in the original code was that the function `findSum` was not defined before it was
called. To fix this, the function definition was added before the function call, allowing the code to execute without any errors.

OUTPUT:

```
PS D:\AI Assisted coding>  & 'C:\Users\vinee\AppData\Local\Microsoft\WindowsApps\python3.12.exe' 'c:\Users\vinee\.vscode\extensions\ms-python.debugpy-2025.19.202602
2301-win32-x64\bundled\libs\debugpy\launcher' '64969' '--' 'D:\AI Assisted coding\10.2-1.py'
15
PS D:\AI Assisted coding>
```

Task Description -3(Code Clarity Improvement)
Task:
Use AI to improve code readability without changing its functionality.
Sample Input Code:
def f(x,y):
return x-y*2
print(f(10,3))
Expected Output-3:
Python code rewritten with meaningful function and variable names,
proper indentation, and improved clarity.

CODE:

```
def f(x,y):
    return x-y
print(f(10,3))
```

The error in the original code was that the function `f` was not defined before it was called. To fix this, the function definition was added before the function call, allowing the code to execute without any errors.

OUTPUT:

```
PS D:\AI Assisted coding>  d:; cd 'd:\AI Assisted coding'; & 'C:\Users\vinee\AppData\Local\Microsoft\WindowsApps\python3.12.exe' 'c:\Users\vinee\.vscode\extensions\
ms-python.debugpy-2025.19.2026022301-win32-x64\bundled\libs\debugpy\launcher' '56399' '--' 'D:\AI Assisted coding\10.2-3.py'
7
```

Task Description -4(Structural Refactoring)
Task:
Use AI to refactor repetitive code into reusable functions.
Sample Input Code:
print("Hello Ram")
print("Hello Sita")
print("Hello Ravi")
Expected Output-4:

Modular Python code using reusable functions to eliminate repetition.

CODE:

```
#ORIGINAL CODE
#print("Hello Ram")
#print("Hello Sita")
#print("Hello Ravi")
#refactor this by creating a reusable function, which improves readability, maintainability, and scalability.
# The original code can be refactored by creating a reusable function called `greet` that takes a name as an argument and prints the greeting message. This way, we can easily greet multiple people without repeating the print statement.
# Here is the refactored code:
def greet(name):
    print(f"Hello {name}")
greet("Ram")
greet("Sita")
greet("Ravi")
```

OUTPUT:

```
PS D:\AI Assisted coding>  d:; cd 'd:\AI Assisted coding'; & 'C:\Users\vinee\AppData\Local\Microsoft\WindowsApps\python3.12.exe' 'c:\Users\vinee\.vscode\extensions\ms-python.debugp
y-2025.19.2026022301-win32-x64\bundled\libs\debugpy\launcher' '51318' '--' 'D:\AI Assisted coding\10.2-4.py'
Hello Ram
Hello Sita
Hello Ravi
```

Task Description -5(Efficiency Enhancement)
Task:
Use AI to optimize Python code for better performance.
Sample Input Code:

```
numbers = [ ]
for i in range(1, 500000):
numbers.append(i * i)
print(len(numbers))
```

Expected Output-5:
Optimized Python code that achieves the same result with improved performance.

CODE:

```
#ORIGINAL CODE
#numbers = []
#for i in range(1, 500000):
 #   numbers.append(i * i)
#print(len(numbers))
#GIVE ME OPTIMIZED FOR THE CODE ABOVE
numbers = [i * i for i in range(1, 500000)]
print(len(numbers))
```

OUTPUT:

```
PS D:\AI Assisted coding>  & 'C:\Users\vinee\AppData\Local\Microsoft\WindowsApps\python3.12.exe' 'c:\Users\vinee\.vscode\extensions\ms-python.debugpy-2025.19.202602
2301-win32-x64\bundled\libs\debugpy\launcher' '64876' '--' 'D:\AI Assisted coding\10.2-4.py'
499999
```